

Guarded Current-Evidence Fusion for Cooperative Multitarget Tracking

Abstract—We propose guarded current-evidence fusion (GCE) for cooperative multitarget tracking with intermittent communication. A robot’s posterior mixes inherited information with its latest local measurement update; observation age alone does not distinguish these contributions. GCE retains a conservative posterior pool and admits a bounded fraction of each current Bernoulli posterior-to-prediction ratio. The same admitted ratio updates the spatial density and the existence normalizer, yielding a closed-form Gaussian implementation. A curvature test rejects unsuitable moment-matched increments, while a zero-vector codec preserves every transmitted floating-point value. We evaluate released two-vehicle V2V4Real detections with a shared score-aware local tracker. On a complete 25-sequence evaluation, GCE obtains OSPA of 3.456 and 3.761 m under reliable and intermittent links, respectively. These are 6.7% and 5.7% below qualified geometric pooling without age, with both paired 95% sequence intervals below zero. Joint spatial and existence correction also improves over a scalar-only correction in both conditions. Missed-target and false-target costs decrease relative to the no-age and recency controls, and common-target localization improves. Exact zero omission reduces fragmented communication relative to the full Gaussian packet, although the resulting payload remains larger than scalar summaries. All real sequences informed development, so these results establish a corpus-level fusion improvement rather than independent generalization.

I. INTRODUCTION

Mobile sensors encounter different targets, occlusions, and communication gaps. When two robots exchange tracks, a recently received posterior can contain old information, whereas a low existence probability can encode a useful current missed detection. A fusion rule must decide which part of each belief to trust without exchanging the entire observation history. This matters in cooperative vehicle tracking, where uncertain target number, independent track births, and brief communication opportunities occur together.

Labeled multi-Bernoulli (LMB) filtering represents a track by its label, existence probability, and conditional spatial density [1]. Geometric pooling, or Kullback–Leibler averaging (KLA), combines such posteriors when cross-correlations are unavailable [2]. A normalized geometric pool retains one common prior in the ideal shared-prior case, but also tempers each sensor’s new likelihood. Changing posterior weights according to observation age does not separately control the inherited and current contributions. Moreover, the existence probability depends on the integral of the spatial product: correcting only existence log-odds generally omits part of the same Bernoulli evidence.

We propose *guarded current-evidence fusion* (GCE). Each robot retains its prediction and local update long enough to encode their Bernoulli ratio. At a communication event,

GCE starts from a conservative pool and supplements it with admitted current ratios. The ratios modify both the spatial distribution and its contribution to existence normalization. Admission depends on current sensing opportunity, association-weighted detection-score support, and whether the moment-matched Gaussian ratio has admissible curvature. Figure 1 presents this continuous information flow.

Separating prior information from likelihood information is established [3], [4]. Our contribution is a guarded Bernoulli realization for approximate local tracking with unequal sensing histories, together with experiments that isolate the spatial correction from the scalar correction and the local detection-score model. We do not assume that the two vehicles’ detector errors or inherited posteriors are independent.

The paper makes three contributions:

- A normalized Bernoulli correction that applies the same admitted current evidence to existence and spatial density, with a tractable Gaussian form and an explicit integrability fallback.
- A bounded admission rule for approximate local updates and a bit-preserving codec that omits exactly zero Gaussian ratios.
- A complete two-condition, 25-sequence comparison with shared score-aware local filtering, external fusion adaptations, component ablations, common-target localization, and actual encoded byte counts.

The main result is an improvement of the complete fusion rule on this development corpus. In particular, the small contribution of conservative history weighting remains uncertain; it is not used to explain all of the observed gain.

II. RELATED WORK

A. Cooperative robotic tracking

Distributed PHD filtering supports robot search and tracking [5], while multirobot reconfiguration addresses changes in sensing resources [6]. LMB representations also support distributed robot landmark mapping [7]. These works establish the broader robotic setting; our experiment keeps sensor trajectories and local dynamics fixed and changes posterior fusion. For cooperative driving, V2V4Real provides real multi-vehicle observations [8], and DMSTrack learns uncertainty within a differentiable multisensor tracker [9]. We use its released per-vehicle detections to study a planar LMB fusion problem, without reproducing that learned tracker or claiming an official 3-D benchmark result.

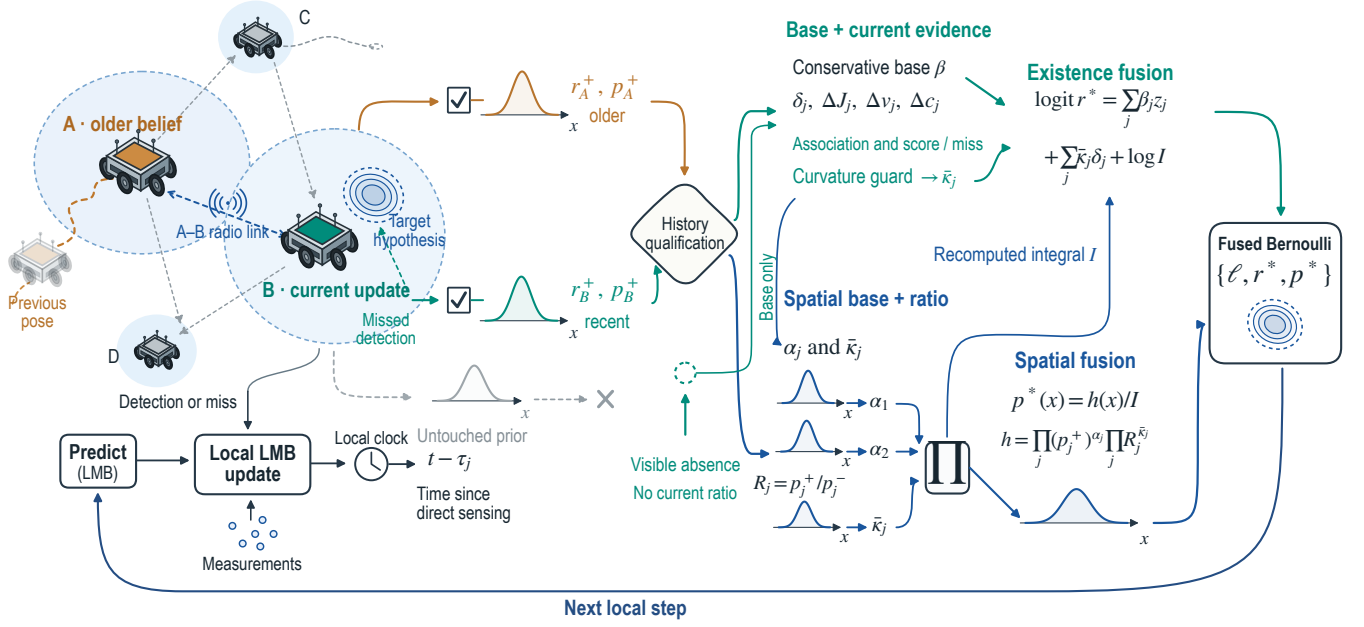


Fig. 1. Guarded current-evidence fusion. Local LMB predictions and updates supply the inherited posterior and current Bernoulli ratio. Qualified support forms the base; association, score or miss evidence, and a curvature test admit each current correction with weight $\bar{\kappa}_j$. The same ratio changes spatial density and existence, coupled by the recomputed integral I . Visible absence contributes only to the base. The fused posterior feeds the next local step. Robot poses and densities are schematic; the real-detection experiment uses two vehicles.

B. Finite-set fusion with different sensing support

Different fields of view require care with absent labels and exclusive information [10], [11]. Minimum-information-loss (MIL) fusion uses arithmetic pooling of labeled finite-set densities [12]. Track consensus instead matches and combines short trajectories, including tracks seen by only some sensors [13]. Our comparisons include an implementation of the public MIL author manuscript [14] and the author's track-consensus functions under the same local detections. Their original experiments and architectures differ from this replay.

Separating cardinality and localization in finite-set fusion is also established: pointwise density consistency alone does not ensure a good cardinality distribution [15]. Heterogeneous fusion can assign confidence at the component level [16]. GCE addresses a narrower question: after choosing an inherited base, how should an admitted local Bernoulli update change both the spatial density and the probability that the target exists?

C. Prior and current likelihood information

Shared-prior Bayesian fusion studies the correction needed when local posteriors contain common prior information [3]. Likelihood consensus aggregates exponential-family likelihood representations across sensors [4]. These principles motivate transmitting current Gaussian natural-parameter increments. GCE uses fractional, source-specific increments because local LMB association and moment projection need not yield an exact independent likelihood. Its curvature guard is an implementation safeguard for that approximation, not a proof that the observation sources are independent.

III. GUARDED CURRENT-EVIDENCE FUSION

A. Local prediction, update, and information support

Each robot predicts its LMB, updates it with the current local measurement set, and exchanges one posterior packet with available neighbors. Fusion output becomes the next local prior. A label is aligned across sources before fusion; births remain local and measurement driven. For a represented label, write the local prediction and update as $f_j^- = (r_j^-, p_j^-)$ and $f_j^+ = (r_j^+, p_j^+)$. The Bernoulli density has empty-set mass $1-r$ and singleton density $rp(x)$. Our replay moment-projects p_j^- and p_j^+ to four-dimensional Gaussians over planar position and velocity.

A direct sensing opportunity occurs when the modeled detection probability at a predicted track is positive. Both a detection and a missed detection advance the local direct-opportunity timestamp τ_j ; receiving another robot's packet does not advance it. A separate history flag records whether a represented belief descends from an informative local update. When informed inputs exist, untouched priors are excluded. An absent label contributes an existence censor only if its source could observe the represented target. Otherwise that source abstains. Censors carry no spatial density. This support handling is shared with the qualified no-age and recency controls.

Let $\mathcal{P}$ be eligible represented sources and $\mathcal{E}$ eligible existence sources, including qualified censors. Normalize ordinary delivered weights w_j separately on the two supports:

$$\alpha_j = \frac{w_j \mathbf{1}[j \in \mathcal{P}]}{\sum_{k \in \mathcal{P}} w_k}, \quad b_j = \frac{w_j \mathbf{1}[j \in \mathcal{E}]}{\sum_{k \in \mathcal{E}} w_k}. \quad (1)$$

All sums below use the corresponding support. A label

without spatial support is omitted. For a represented source, define

$$a_j = \rho + (1 - \rho)e^{-(t-\tau_j)\Delta t/\lambda}, \quad q_j = \frac{b_j a_j}{\sum_k b_k a_k}, \quad (2)$$

with $a_j = \rho$ before any local opportunity and $a_j = 1$ for a censor. We use $\rho = 0.25$ and $\lambda = 5$ s.

B. A conservative inherited base

Set $z_j = \text{logit}(r_j^+)$ and choose the existence weights

$$\beta = \begin{cases} q, & \sum_j (q_j - b_j) z_j < 0, \\ b, & \text{otherwise.} \end{cases} \quad (3)$$

Thus history can lower the base existence log-odds but cannot raise them. The spatial base retains the ordinary weights α . An unnormalized base has masses

$$A_0 = \prod_j (1 - r_j^+)^{\beta_j}, \quad (4)$$

$$A_1(x) = \prod_j (r_j^+)^{\beta_j} \prod_j p_j^+(x)^{\alpha_j}. \quad (5)$$

These are separate existence and spatial weights; when the weights and supports coincide, the base reduces to ordinary Bernoulli KLA. The no-history ablation sets $\beta = b$ throughout. Its purpose is to test the small conservative history adjustment separately from the current evidence correction.

C. Admitting current positive and negative evidence

The local existence increment is

$$\delta_j = \text{logit}(r_j^+) - \text{logit}(r_j^-). \quad (6)$$

The local tracker incorporates a detection score s through a fitted positive-to-negative mark ratio $L(s) > 0$. Let W_{jm} be the normalized local association mass for assigning this Bernoulli to measurement m , and let $A_j = \sum_m W_{jm}$. For a current sensing opportunity, use

$$g_j^+ = \sum_m W_{jm} \max\left(0, \frac{L(s_m) - 1}{L(s_m) + 1}\right), \quad (7)$$

$$g_j^- = (1 - A_j) \frac{p_{D,j}}{2 - p_{D,j}}. \quad (8)$$

Both gates are zero without a current opportunity. The first admits positive increments in proportion to association-weighted mark discrimination; the second weights the missed-detection branch by its bounded likelihood contrast. They satisfy $g_j^+ \leq A_j$, $g_j^- \leq 1 - A_j$, and hence $g_j^+ + g_j^- \leq 1$. They are bounded admission scores, not calibrated posterior error bounds.

If at least two eligible sources represent the label and none of the eligible existence inputs is a censor, define

$$\kappa_j = (1 - \beta_j) \begin{cases} g_j^+, & \delta_j \geq 0, \\ g_j^-, & \delta_j < 0. \end{cases} \quad (9)$$

Otherwise set every $\kappa_j = 0$. This restriction avoids constructing an unavailable current ratio for an absent label. The no-score ablation replaces the contrast in (7) by one, retaining the same score-aware local tracker and setting $g_j^+ = A_j$.

D. Joint existence and spatial normalization

For each admitted source, multiply the same fractional Bernoulli ratio $(f_j^+/f_j^-)^{\kappa_j}$ into both hypotheses in (5). The resulting spatial product is

$$h(x) = \prod_j p_j^+(x)^{\alpha_j} \prod_j \left[\frac{p_j^+(x)}{p_j^-(x)} \right]^{\kappa_j}, \quad I = \int h(x) dx. \quad (10)$$

Proposition 1 (normalized Bernoulli form). For $0 < r_j^-, r_j^+ < 1$ and $0 < I < \infty$, normalizing these empty and singleton masses yields

$$p^*(x) = h(x)/I, \quad (11)$$

$$\text{logit}(r^*) = \sum_j \beta_j z_j + \sum_j \kappa_j \delta_j + \log I. \quad (12)$$

Proof. The empty mass is $H_0 = A_0 \prod_j [(1 - r_j^+)/(1 - r_j^-)]^{\kappa_j}$. The integrated singleton mass is $H_1 = I \prod_j (r_j^+)^{\beta_j} \prod_j (r_j^+/r_j^-)^{\kappa_j}$. Taking $\log(H_1/H_0)$ gives (12); normalizing the singleton shape gives (11). $\square$

A scalar-only correction keeps the first two terms in (12) but uses the old spatial pool and its integral. It therefore implements a different approximation from the full admitted Bernoulli ratios. We evaluate this scalar rule as a complete recursive component control, in addition to substitutions on fixed fusion inputs.

In the special case of a common prior f^- , equal weights $\alpha = \beta$, unit gates, and independent exact local observations, $\kappa_j = 1 - \beta_j$. The full product then reduces to $\prod_j f_j^+/(f^-)^{M-1}$, the standard shared-prior Bayesian combination [3]. This algebraic check does not make approximate LMB associations or correlated detector errors exact.

E. Gaussian correction and curvature guard

For $p = \mathcal{N}(m, P)$, write

$$\log p(x) = -\frac{1}{2} x^\top J x + v^\top x + c, \quad J = P^{-1}, \quad v = Jm, \quad (13)$$

where $c = -[d \log(2\pi) + \log |P| + m^\top Jm]/2$ and $d = 4$. A source transmits the differences $(\Delta J_j, \Delta v_j, \Delta c_j)$ between its update and prediction. The symmetric precision difference needs ten numbers, the information vector four, and the constant one.

Moment projection can widen a covariance after ambiguous association, so this Gaussian ratio need not be log-concave. GCE rejects the entire source correction, including its scalar increment, when

$$\lambda_{\min}(\Delta J_j) < -10^{-10} \max(1, \|J_j^+\|_2, \|J_j^-\|_2). \quad (14)$$

Let $\bar{\kappa}_j$ be the resulting weight. No eigenvalue is clipped. The aggregate natural parameters are

$$J = \sum_j \alpha_j J_j^+ + \sum_j \bar{\kappa}_j \Delta J_j, \quad (15)$$

$$v = \sum_j \alpha_j v_j^+ + \sum_j \bar{\kappa}_j \Delta v_j, \quad (16)$$

$$c = \sum_j \alpha_j c_j^+ + \sum_j \bar{\kappa}_j \Delta c_j. \quad (17)$$

For positive definite J , the normalized quantities are

$$P^* = J^{-1}, \quad m^* = J^{-1}v, \quad (18)$$

$$\log I = c + \frac{1}{2}[d \log(2\pi) - \log |J| + v^\top J^{-1}v]. \quad (19)$$

Use $\bar{\kappa}$ in (12) as well. A nonfinite aggregate, failed Cholesky factorization, or reciprocal condition estimate below 10^{-12} discards all residual terms and restores the base. When no residual is admitted, the original base implementation is retained exactly. In exact arithmetic, positive semidefinite admitted precision gains and a positive definite base ensure a positive definite aggregate; the final test also covers numerical failures. The no-curvature ablation removes only (14), retaining this necessary integrability test.

Existence log-odds use a $[10^{-9}, 1 - 10^{-9}]$ numerical clip, and differences within 10^{-12} do not activate the history switch. Current gates and ratios are cleared after fusion; the next packet recomputes them from that robot's next local update. For state dimension d and M eligible sources, the dense Gaussian arithmetic costs $O(Md^3)$ and adds $O(d^2)$ numbers per track.

F. Exact zero-vector transmission

The full Gaussian implementation extends a 232-byte scalar Bernoulli summary by fifteen 64-bit numbers, giving 352 bytes per track plus a 32-byte packet header. Many local updates leave all fifteen differences exactly zero. A two-byte tag distinguishes a full vector from an omitted zero vector and preserves the sign bit of each omitted zero. All nonzero vectors retain their original 120 bytes. For n tracks and k nonzero vectors, the actual packet length is

$$B = 32 + 234n + 120k \quad \text{bytes.} \quad (20)$$

This codec has no threshold or quantization parameter. Decoded numerical values, and hence the fusion rule, are unchanged. We verify complete recursive trajectories with the codec rather than estimating its savings from the mean number of tracks alone.

IV. EXPERIMENTS

A. Real detections, sequence scope, and local model

We use the released per-vehicle, no-fusion detections from DMSTrack [9] on V2V4Real [8]. The development subset contains all nine released validation sequences (1,993 paired frames). The main evaluation uses 25 train sequences (5,601 paired frames), excluding the seven train IDs previously used in initial transfer checks. Each sequence is complete. The released detector was trained on the train split. All 34 reported sequences informed method development, so this is a fusion development evaluation and not an independent detector or generalization test.

Inputs include each vehicle's detections, detection scores, per-frame relative transforms, and car annotations. Released detections already use a confidence cutoff of 0.20; we apply no further score threshold. We evaluate planar car centers within 40 m of either sensor and inside the annotated ego-frame rectangle $[-70.4, 70.4] \times [-40, 40]$ m, excluding a 3

m disk around each vehicle. The same domain is applied to estimates and truth, and each local sensor uses its own 40 m range. This range models sensing support rather than occlusion. Coordinates follow the current ego frame; the constant-velocity model does not compensate ego motion.

Every method in the main comparison uses the same score-aware LBP-LMB local update. The model uses a 0.1 s scan interval, survival probability 0.99, white-acceleration intensity $25 \text{ m}^2/\text{s}^3$, position observation covariance $I_2 \text{ m}^2$, in-domain $p_D = 0.9$, and nominal Poisson clutter mean three per sensor scan. The clutter density is $3/(\pi 40^2)$. Each sensor independently creates a birth at every previous-scan detection with existence 0.01, zero velocity, and covariance $\text{diag}(16, 16, 225, 225)$. Bernoullis with $r \leq 0.001$ are pruned, retained mixtures are moment-projected to one Gaussian, and MAP-cardinality extraction produces the reported set.

Observation-history qualification is identical across the geometric density controls. In the single-Gaussian implementation, an absent label is eligible for an existence censor when its represented mean is inside the absent source's modeled FoV and expected detection is at least 0.1. The censor's existence bound is the transmission threshold 0.001. Missing spatial densities do not participate in the spatial product.

B. Common detection-score model

Calibration labels are obtained by one-to-one, per-sensor geometric assignment of detections to truth within 12 m. We fit $u(s) = \sigma(a \logit s + b)$ with $a \geq 0$ by minimizing sequence-balanced binary log loss plus $0.001a^2$. Each development sequence uses a fit on the other eight. For the 25-sequence evaluation, the fit uses all nine development sequences and remains fixed: $a = 2.8858$, $b = 2.1564$. Ground truth is used for this offline fit and evaluation; online births, association, and fusion use detections and communicated beliefs.

The local detection-association likelihood is multiplied once by

$$L(s) = \frac{u(s)}{1 - u(s)} \frac{1 - \pi}{\pi}, \quad (21)$$

where π is the sequence-balanced positive fraction of the fitting data. Probabilities are clipped to $[10^{-6}, 1 - 10^{-6}]$ before this conversion. Treating the resulting score ratio as a separable mark likelihood is an approximation. The main baselines receive exactly the same ratios, so their comparison with GCE does not attribute local calibration gains to the fusion correction.

C. Communication and comparison methods

Reliable links deliver both directed packets at every scan. Intermittent links use independent directed loss of 0.1 and a common bidirectional outage over the middle 20% of scans. Every method shares the same delivery realization and one packet opportunity per direction per scan. A receiver with a delivered neighbor uses weights $[0.5, 0.5]$; otherwise it retains its own local update. Geometry and poses are supplied inputs. Byte totals include a modeled 256-byte control charge

per scan and 16,384-byte packet fragments. The loss trace does not depend on packet size.

No-age KLA uses the qualified spatial and existence weights α, b . *Recency* replaces b with q in existence only. *Conservative* uses β from (3) without a current correction. The internal *Capped-score* and *Capped-cal.* controls bound recency existence by $\min(r_{\text{rec}}, \max(r_{\text{noage}}, U))$, where U is the largest current association-weighted raw or calibrated score among upweighted sources with input existence at least 0.5. These controls preserve the ordinary spatial pool. *Scalar* retains GCE's base and positive and negative gates but corrects only existence log-odds, leaving the old spatial density and normalizer.

MIL-AM implements the public author manuscript's augmented label assignment and common/exclusive label-subspace fusion [14]. Gaussian labels are assigned by symmetric KL cost with unmatched cost 50 per side; the same alignment is used by the geometric density methods. Common labels use arithmetic existence pooling and existence-weighted spatial mixtures, and exclusive labels retain their source density. This is an adaptation of the public manuscript, not a verified reproduction of the complete later TAES experiment [11].

TC-5/TC-10 call the author's kinematic track matching and two-stage multi-neighbor fusion functions [13], with five- or ten-scan histories, matching order one, and cutoff 12 m. They use independently labeled tracks from the same local backend. As in that architecture, the fused track output does not feed back into local densities. We evaluate set estimates without network-wide label reconciliation. *Local* uses the shared backend without communication.

D. Metrics and component checks

The primary metric is position OSPA with $p = 2$ and cutoff 12 m [17]. We also report GOSPA with $p = 2, c = 12$ m, $\alpha = 2$ [18]; its missed and false components are *squared costs* in m^2 . For localization, each method is assigned to truth separately, then errors are pooled on robot-time-truth instances matched by both methods. Common support and RMSE are reported together.

Sequences receive equal weight. Paired 95% percentile intervals use 10,000 resamples of complete sequences, with no multiplicity adjustment. Frames and vehicles are not independent replicates, and the intervals are descriptive because the trajectories have informed development. Every declared sequence and both link conditions are retained.

All three GCE ablations run complete recursive trajectories: no curvature guard, no conservative history switch, and no positive score constraint. Single-Bernoulli checks cover the centralized shared-prior identity, conflicting observations, zero gates, single-source behavior, and nonintegrable aggregates. A separate Python implementation reconstructs local Gaussian ratios and the fused existence, mean, and covariance, and recomputes trajectory metrics and packet counts. Finally, the zero-vector codec is replayed on all 34 primary trajectories under both link conditions, requiring identical tracking outputs and original local and fusion diagnostics.

V. RESULTS

A. Main set-estimation comparison

Table I reports all 25 sequences with the same score-aware local tracker. GCE attains OSPA of 3.456 m under reliable links and 3.761 m under intermittent links. Relative to No-age KLA, the reductions are 6.7% and 5.7%; relative to Recency they are 5.9% and 4.8%. GCE improves on No-age KLA in 20/25 reliable sequences and 24/25 intermittent sequences. Against Recency, it improves in 17/25 and 18/25 sequences, respectively.

The paired GCE-minus-No-age OSPA differences are -0.246 m [-0.388, -0.112] and -0.227 m [-0.331, -0.143]. The corresponding differences from Recency are -0.216 m [-0.345, -0.089] and -0.190 m [-0.294, -0.092]. All four intervals lie below zero. The intervals against Conservative and both capped controls also lie below zero in both conditions. Figure 2 retains all sequence differences for these controls and the Scalar component.

Both missed-target and false-target squared costs decrease relative to No-age KLA and Recency. Against No-age KLA, mean missed cost decreases by 16.9%/13.6%, and false cost by 10.8%/6.3% (reliable/intermittent). On common matched truth, GCE/No-age RMSE is 0.907/0.964 m with reliable links and 1.050/1.085 m with intermittent links, on 80 620 and 79 309 instances. The lower set error therefore does not arise solely from trading worse localization for a different target count.

GCE also has lower sequence-macro OSPA than MIL-AM and both TC windows in both conditions. These are comparisons with the stated adaptations; TC's separate local-track architecture and MIL's arithmetic pooling differ from recursive geometric density fusion. The controlled No-age, Recency, and Scalar comparisons more directly isolate the change to the fusion rule.

B. The contribution of joint normalization

The complete Scalar control obtains 3.526/3.832 m OSPA. The paired GCE-minus-Scalar differences are -0.070 m [-0.135, -0.013] and -0.070 m [-0.129, -0.021]. Here differences retain the GCE-minus-control convention; both intervals lie below zero. GCE also lowers common-target RMSE from 0.944 to 0.906 m and from 1.082 to 1.055 m. Relative to Scalar, the correction reduces mean missed cost while slightly increasing mean false cost, as shown in Table II.

To distinguish immediate fusion effects from changes accumulated through recursive filtering, we also re-extract estimates on GCE's saved fusion inputs. Using the old spatial pool and Scalar existence gives OSPA 3.475/3.535 m on actual fusion frames, versus 3.456/3.510 m for the joint correction. Replacing only the mean or only the existence gives intermediate diagnostics. These substitutions do not reproduce the Scalar trajectory: the larger full-run difference also includes later association, pruning, and extraction effects.

TABLE I

MAIN EVALUATION OVER 25 COMPLETE SEQUENCES. ALL METHODS USE THE SAME SCORE-AWARE LOCAL TRACKER. OSPA IS IN M; MISSED AND FALSE GOSPA COMPONENTS ARE SQUARED COSTS IN m^2 . LOWER IS BETTER.

Method	Reliable links			Intermittent links		
	OSPA	Missed	False	OSPA	Missed	False
Local	5.914	190.030	45.556	5.914	190.030	45.556
MIL-AM	4.014	74.566	56.973	4.163	82.428	58.363
TC-5	4.253	68.620	75.289	4.693	101.053	64.733
TC-10	4.414	88.338	65.001	4.829	116.571	57.691
No-age KLA	3.702	85.564	40.715	3.988	92.977	44.776
Recency	3.672	78.330	43.986	3.952	87.113	47.603
Conservative	3.672	86.046	37.436	3.956	93.865	42.483
Capped-score	3.666	83.065	39.074	3.913	90.189	43.663
Capped-cal.	3.650	80.776	40.418	3.897	88.391	44.318
Scalar	3.526	78.625	34.536	3.832	86.342	40.523
GCE	3.456	71.095	36.303	3.761	80.344	41.971

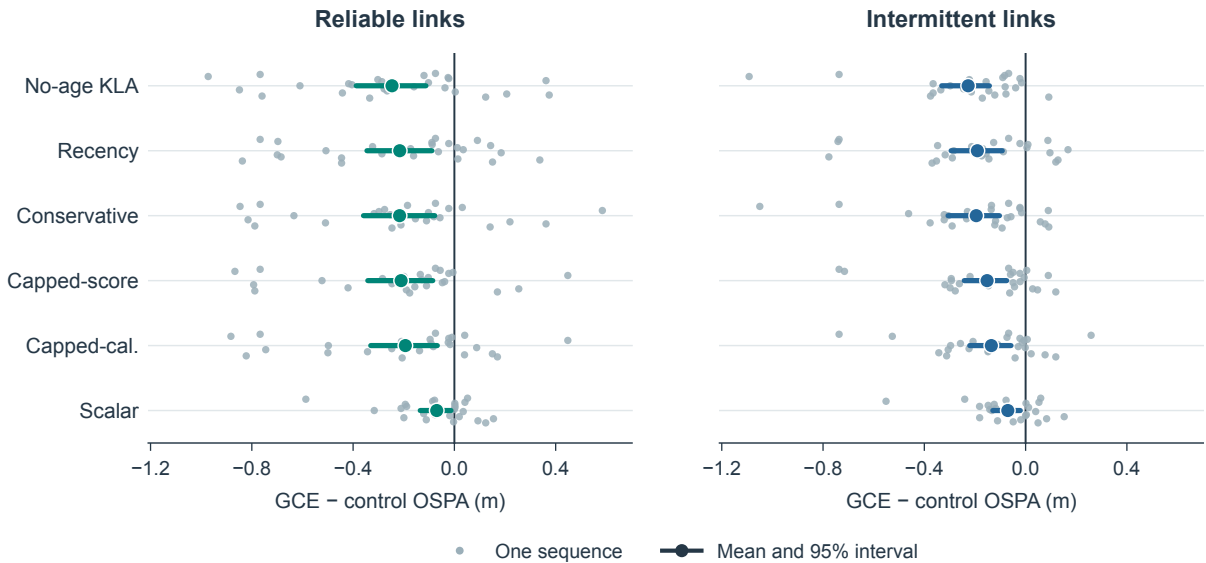


Fig. 2. Complete 25-sequence comparisons. Small dots are individual sequence OSPA differences, GCE minus the labeled control; large points and bars show equal-sequence means and 95% percentile intervals from 10,000 sequence resamples. Negative values favor GCE. All points and both link conditions are included; intervals describe this development corpus.

C. What the guards and history term contribute

Figure 3 and Table II report every declared component removal. Omitting the curvature guard raises mean OSPA in both conditions; its paired interval excludes zero only with intermittent links. The unguarded version encounters two nonintegrable aggregates under reliable links, which trigger the shared fallback. GCE encounters none. The largest observed immediate position changes are 6.85/7.93 m for GCE, versus 28.68/49.62 m without the source guard. These observed maxima illustrate the effect of rejecting unsuitable Gaussian ratios; they are not error bounds.

Removing the conservative history switch changes mean OSPA by only about 0.01 m, and both intervals include zero. Removing the positive score constraint substantially increases mean false-target cost but reduces missed-target cost. Its total OSPA intervals also include zero. Thus the current

experiment supports the full rule and the joint correction over Scalar, while leaving some individual design choices only weakly resolved.

D. Development results and the common local model

On the nine-sequence development subset, GCE attains 3.877/4.206 m OSPA, compared with 4.051/4.312 m for No-age KLA and 3.928/4.234 m for Scalar. All fixed methods and both conditions are retained in the accompanying source data. The development Scalar comparison is weaker than the 25-sequence result: its two intervals include zero.

The score-aware local model itself matters. In the 25-sequence evaluation, unmarked No-age KLA has OSPA 4.217/4.413 m, compared with 3.702/3.988 m when both methods use the common score-aware update. That gain belongs to the local observation model. The additional

TABLE II

COMPLETE RECURSIVE COMPONENT COMPARISONS ON THE SAME 25 SEQUENCES. THE SCORE CONSTRAINT ABLATION RETAINS THE COMMON SCORE-AWARE LOCAL UPDATE. UNITS MATCH TABLE I.

Method	Reliable links			Intermittent links		
	OSPA	Missed	False	OSPA	Missed	False
GCE	3.456	71.095	36.303	3.761	80.344	41.971
Scalar	3.526	78.625	34.536	3.832	86.342	40.523
No curvature guard	3.491	74.845	35.376	3.805	84.047	41.615
No history switch	3.467	70.282	38.148	3.776	80.266	43.105
No positive score constraint	3.713	55.993	59.014	3.922	69.042	57.784

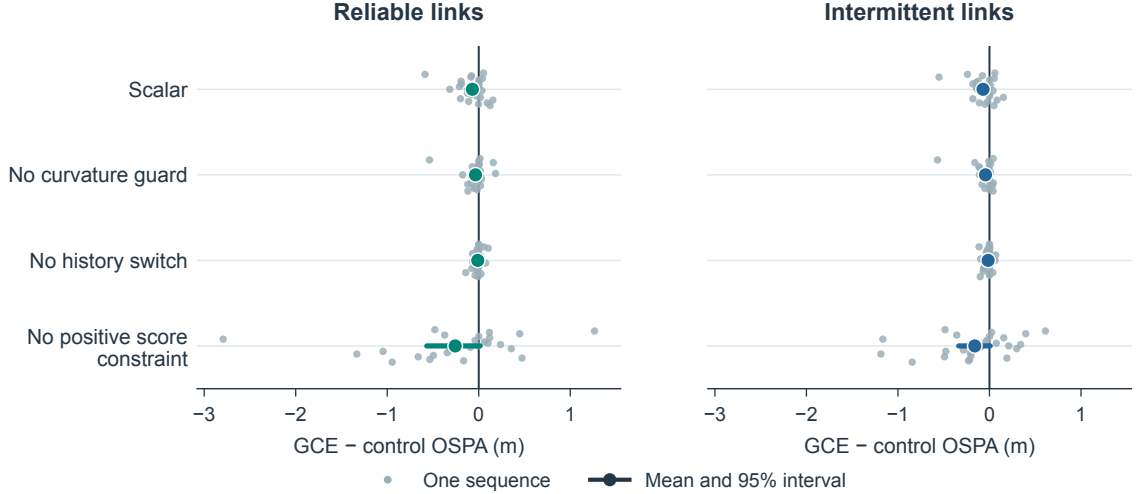


Fig. 3. Component evidence on all 25 sequences. GCE-minus-control OSPA differences use the same point and interval conventions as Fig. 2. The Scalar control removes the spatial ratio; the other controls remove one admission or base-weighting component. Intervals crossing zero do not establish an OSPA contribution for that component.

GCE improvement is measured against the latter, and the no-positive-score-constraint ablation retains that same local model.

E. Actual communication cost

Table III and Fig. 4 use the native zero-vector codec for GCE. It reproduces every primary trajectory and all original local and fusion diagnostics under both link conditions. Across the 25 sequences, raw payload falls by 20.3%/21.1% relative to the full 352-byte Gaussian representation; fragmented totals fall by 11.3%/10.8%.

Compared with No-age KLA, GCE's compressed raw payload is still 17.3%/21.5% larger, while fragmented cost is 4.1%/7.0% larger. The gain therefore trades additional transmitted information for lower tracking error. The 16,384-byte fragmentation model makes byte savings discontinuous at packet boundaries. We do not infer real-network latency, throughput, or size-dependent loss from this fixed delivery trace.

VI. DISCUSSION AND CONCLUSION

GCE separates a conservative inherited pool from admitted current local updates, then normalizes their Bernoulli product jointly. The Gaussian form makes explicit why a spatial

ratio also changes existence: the integral of the corrected spatial product is part of the singleton mass. The complete 25-sequence result improves on qualified no-age pooling, recency weighting, conservative and capped controls, and the Scalar component in both link conditions. Relative to the two original geometric controls, missed cost, false cost, and common-target localization improve together.

The ablations narrow the interpretation. The full spatial correction has a supported advantage over Scalar in the main corpus. The curvature guard prevents two invalid aggregates observed in its removal, but its OSPA effect is not resolved in both conditions. The history switch has a small, uncertain contribution. The score constraint exchanges fewer false targets for more misses, so its merit cannot be inferred from false cost alone. These distinctions favor studying complete Bernoulli evidence over attributing the entire result to observation age.

Three limits remain. First, current ratios come from approximate association and Gaussian moment projection; a positive curvature check does not establish source independence or remove all inherited correlation. Second, the fitted mark model depends on geometric calibration labels and a separability approximation. All real sequences have informed

TABLE III

MEAN ACTUAL MiB PER SEQUENCE OVER 25 SEQUENCES. GCE USES THE EXACT ZERO-VECTOR CODEC. FRAGMENTED TOTALS INCLUDE ATTEMPTED PACKET FRAGMENTS AND MODELED CONTROL TRAFFIC; DELIVERED RAW BYTES COUNT SUCCESSFUL DELIVERIES ONLY.

Method	Reliable links			Intermittent links		
	Raw	Delivered	Fragmented	Raw	Delivered	Fragmented
Local	0.000	0.000	0.000	0.000	0.000	0.000
MIL-AM	6.241	6.241	10.205	5.858	4.163	10.020
TC-5	1.136	1.136	7.056	1.136	0.779	7.056
TC-10	2.223	2.223	7.234	2.223	1.521	7.234
No-age KLA	3.890	3.890	8.347	3.882	2.748	8.294
Recency	3.591	3.591	8.139	3.650	2.575	8.103
Conservative	3.686	3.686	8.228	3.717	2.630	8.248
Capped-score	3.803	3.803	8.283	3.810	2.690	8.286
Capped-cal.	3.745	3.745	8.251	3.790	2.673	8.253
Scalar	3.727	3.727	8.190	3.886	2.738	8.322
GCE	4.562	4.562	8.687	4.717	3.338	8.878
GCE full Gaussian packet	5.724	5.724	9.788	5.978	4.219	9.951

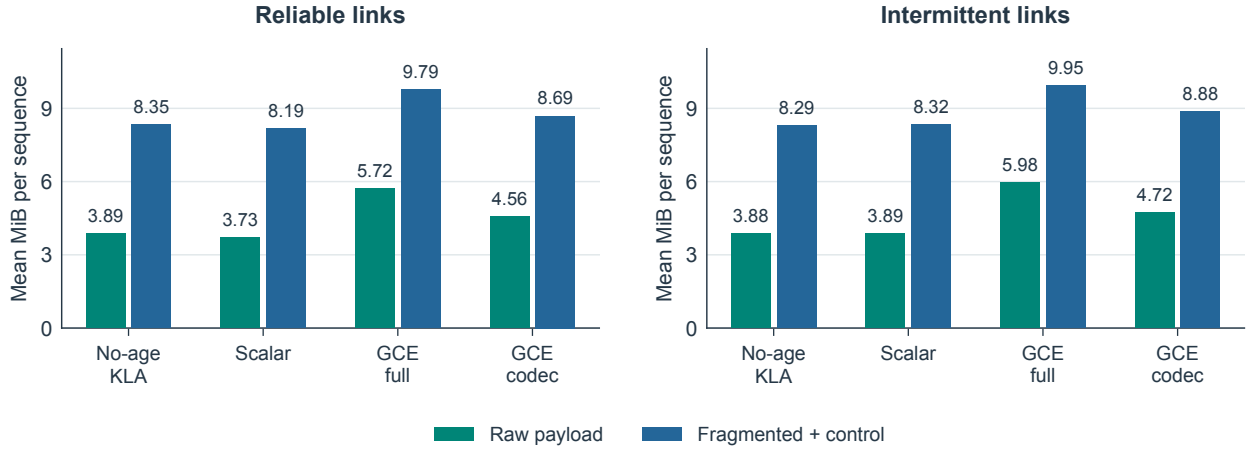


Fig. 4. Actual mean MiB per sequence over 25 sequences. Raw payload and fragmented totals compare No-age KLA, Scalar, and two GCE packet formats. Full Gaussian and exact zero-vector GCE packets yield identical tracking trajectories. Fragmented totals include modeled control traffic and use 16,384-byte fragments. The delivery trace is fixed across methods.

development, and the main detections come from the detector’s training split. A new route and detector split is needed to test generalization. Third, the replay is planar, involves two vehicles, uses uncompensated moving ego coordinates, and emulates communication. Larger robot networks, pose uncertainty, and measured radio effects remain outside this experiment.

Within that scope, the result establishes a concrete fusion improvement: bounded current evidence can be restored while keeping existence and spatial normalization consistent. Exact zero omission recovers part of the extra communication cost without changing a track or a probability.

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